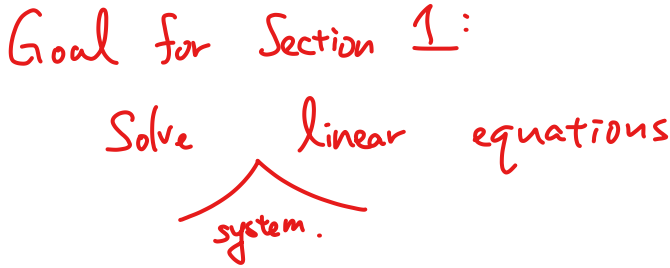
MA1522 Linear Algebra for Computing

Lesson 1: Linear Systems, Row Operations and Row-Echelon Forms



Pollev



Basic rules:

- Please (mainly) ask the questions related to the course.

Section 1.1 Linear Equations

A **linear equation** in the variables $x_1, \ldots, x_n$ is

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

standard form

with constant **coefficients** $a_1, \ldots, a_n$ and constant term b. It is **homogeneous** when $b = 0$.

Every variable appears to the first power only: no x_i^2, x_ix_j, $1/x_i$, $\sqrt{x_i}$, $\sin x_i$, ...

Example 1

Which of these are linear equations? If so, write it in standard form and say whether it is homogeneous.

1. $z - 2x - 2y = 7$ standard form
2. $y = x + 1$ $-x + y = 1$
3. $x(y + z) = 3$ $xy + xz = 3$
4. $2x = 3y$ $2x - 3y = 0$ homogeneous.
5. $z = 2(x + y)$ $-2x - 2y + 2 = 0$ homogeneous.
6. $y = (2x + 3)^2$ $y = 4x^2 + 12x + 9$

Section 1.1 Linear Systems

A **linear system** is a set of m linear equations in the same n variables. It is encoded by its **augmented matrix**:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases} \longleftrightarrow \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right)$$

- A **solution** satisfies every equation at once; the system is **consistent** if at least one exists, and the **general solution** describes all of them.
- The system is **homogeneous** if $b_1 = \cdots = b_m = 0$. A homogeneous system is always consistent: $\mathbf{0}$ is the **trivial solution**.
- Exactly one of three things happens: no solution, exactly one solution, or infinitely many.

Question 1

Consider the linear system $\begin{cases} x + 2y - z = 1 \\ y - z = 3 \end{cases}$.

Explain why

$$x = -5 - t, \quad y = 3 + t, \quad z = t, \qquad t \in \mathbb{R}$$

is a general solution of this system.

There are **two** things to check: every listed triple is a solution, and every solution appears in the list.

Question 1 — solution

$$\begin{cases} x + 2y - z = 1 \\ y - z = 3. \end{cases} \qquad \begin{cases} x = -5 - t \\ y = 3 + t \\ z = t \end{cases} \qquad t \in \mathbb{R}.$$

Step 1 Verify those are all sol.

$(-5-t) + 2(3+t) - t \stackrel{?}{=} 1$ correct by direct computation

$(3+t) - t \stackrel{?}{=} 3$ ✓.

$a = -5-c$

$b = 3+c.$

$c = c$

Step 2 It contains all solutions

Suppose $(a, b, \underbrace{c}_{=t})$.

WTS: $a = -5-t$ ✓, $b = 3+t$ ✓

(want to show)

$y - z = 3 \Rightarrow b - t = 3 \Rightarrow b = t+3$

$x + 2y - z = 1 \Rightarrow a + 2(t+3) - t = 1$

$\Rightarrow a = -t - 5$ □.

Challenge

Explain why

$$x = -2 - s, \quad y = s, \quad z = s - 3, \qquad s \in \mathbb{R}$$

$x = -s - t \quad y = 3+t \quad z = t \qquad t \in \mathbb{R}$.

is *also* a general solution of the same system $\begin{cases} x + 2y - z = 1 \\ y - z = 3 \end{cases}$.

Remark

The form in which a general solution is written is not unique. The same set of solutions can be described by different parametrisations, so two answers that look nothing alike may both be correct.

Question: Given two expressions of general solutions, how to show that they are the same or not?

Section 1.2 Elementary Row Operations

The three **elementary row operations** on a matrix:

- **Interchange** two rows: $R_i \leftrightarrow R_j \quad (i \neq j)$.
- **Multiply** a row by a nonzero constant: $aR_i \quad (a \neq 0)$.
- **Add** a multiple of a row to another: $R_i + cR_j \quad (i \neq j)$.

Every elementary row operation is **reversible**.

Section 1.2 Row Equivalence

A and B are **row equivalent** if one is obtained from the other by a series of elementary row operations,

$$A = A_0 \to A_1 \to \cdots \to A_k = B.$$

By reversibility the relation is symmetric.

Theorem

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

This licenses the whole method: replace a system by a simpler row equivalent one, without losing or gaining any solution.

Question 2

Write the system as an augmented matrix, carry out the given row operations, and read off the solution.

$$\begin{cases} 2x + 3y = 12 \\ x + 2y = 7 \end{cases}$$

$R_1 \leftrightarrow R_2$ / $R_2 - 2R_1$ / $-R_2$.

$$\left(\begin{array}{cc|c} 2 & 3 & 12 \\ 1 & 2 & 7 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 2 & 7 \\ 2 & 3 & 12 \end{array}\right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -1 & -2 \end{array}\right)$$

$-2 \quad -4 \quad -14.$

$\xrightarrow{-R_2}$

$$\left(\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 2 \end{array}\right)$$

$$\begin{cases} x + 2y = 7 \\ y = 2. \end{cases} \Rightarrow \begin{array}{l} y = 2 \\ x = 7 - 2y = 3 \end{array}$$

$2 \cdot 2.$

Section 1.3 Row-Echelon Forms

A **zero row** has all entries 0. The **leading entry** of a nonzero row is its leftmost nonzero entry.

A matrix is in **row-echelon form** (REF) if

1. all zero rows are grouped at the bottom;
2. each leading entry is strictly to the right of the leading entry of the row above it.

It is in **reduced row-echelon form** (RREF) if moreover

3. every leading entry equals 1;
4. every column containing a leading entry has all its other entries 0.

A column containing a leading entry is a **pivot column**; the others are **non-pivot columns**.

A matrix has *many* REFs but exactly *one* RREF — hence "*a* REF of A" but "*the* RREF of A".

Example 2

For each matrix, state whether it is in REF, in RREF, or neither.

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}$ REF

not RREF

(b) $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$ not REF.

(c) $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ not REF

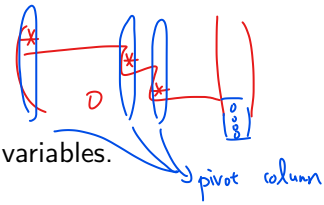
(d) $\begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ REF

✓ RREF

(e) $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ REF.

not RREF.

Section 1.3 Reading solutions off a REF



Let R be a REF of the augmented matrix of a system in n variables.

- If the rightmost column of R is a pivot column, some row reads $0 = b$ with $b \neq 0$: the system is **inconsistent.**

- Otherwise the system is **consistent**:
  - non-pivot columns $\rightarrow$ arbitrary **parameters**;
  - pivot columns $\rightarrow$ solved by **back substitution**, from the bottom row upwards.

Number of parameters = number of non-pivot columns among the first n columns. The solution is unique exactly when all n are pivot.

Question 3

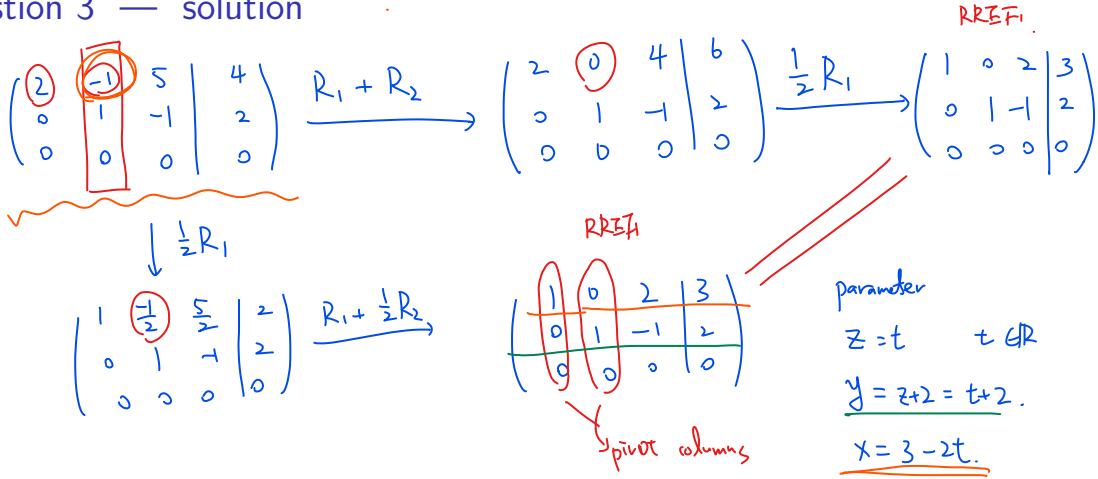
The matrix

$$\left(\begin{array}{ccc|c} 2 & -1 & 5 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

is in REF but not in RREF.

Transform it into its RREF using elementary row operations, and solve the corresponding system.

Question 3 — solution



Next lessons

Today : no submission needed.

Friday : submission deadline. 10:30.

More details will be announced.

- **Section 1.4 Gaussian Elimination**
- **Section 1.5 More on Linear Systems**